Journal Name

ARTICLE TYPE

Cite this: DOI: 00.0000/xxxxxxxxxx

Can PFAS adsorbents be predictively identified through machine learning? A study of PFOA-targeted screening in metalorganic frameworks†

M. Hasan Mansoor,*[a] Sousa Javan Nikkah,[b‡] Mosab Bazargani,[a‡] and Soumya Mukherjee[b]

Received Date
Accepted Date

DOI: 00.0000/xxxxxxxxxx

The freshwater and drinking-water PFAS problem; Why PFOA is used as a prototype contaminant; Why database-led predictive discovery is needed; Core workflow: labelled literature set + unlabelled CoRE-MOF/CSD discovery space + simulation-informed ML refinement; Key outcome: predictive identification of chemically plausible benchmark adsorbents; General significance for PFAS sorbent discovery. Any references in the abstract should be written out in full *e.g.* [Surname *et al.*, *Journal Title*, 2000, **35**, 3523].

1 Introduction

1. PFAS contamination as a freshwater and drinking-water challenge

2. Why current sorbent discovery remains too slow and too empirical

3. PFOA as a chemically and regulatorily meaningful prototype among PFAS

4. Why metal–organic frameworks are attractive as tunable sorbents

5. Why the Cambridge Structural Database and CoRE-MOF subsets create a discovery opportunity

6. Why predictive design is still missing for PFAS-active metal–organic sorbents

7. Study hypothesis and paper objective

Rough draft: Per- and polyfluoroalkyl substances (PFAS) have become pervasive contaminants of freshwater and drinking-water supplies. Their widespread use in industrial processes and consumer products, coupled with the exceptional stability of their fluorinated carbon chains, has led to their persistence across aquatic environments. Continued exposure to several PFAS has been associated with adverse health outcomes, while increasingly stringent drinking-water limits require their removal at concentrations approaching the ngL^{-1} regime. These demands present a formidable separation challenge: PFAS must be selectively scavenged from chemically complex aqueous matrices despite their low concentrations, diverse chain lengths and amphiphilic character.

Adsorption remains the most readily deployable technology for PFAS removal. Activated carbons are widely used, but their performance can be limited by slow uptake, competition from dissolved organic matter, poor affinity for shorter-chain PFAS and energy-intensive regeneration. These shortcomings have motivated the search for regenerable sorbents whose chemical environments can be tailored to recognise PFAS selectively. Metalorganic frameworks (MOFs) are particularly attractive in this regard because their pore dimensions, surface chemistry, hydrophobicity, charge distribution and metalligand composition can, in principle, be controlled independently. Recent studies have further shown that PFAS uptake cannot be rationalised by accessible surface area or pore volume alone. Rather, the interplay between confinement, hydrophobic association and electrostatic interactions appears to govern adsorption, highlighting the need for a multidimensional approach to sorbent design.

The chemical diversity that makes MOFs promising also makes their discovery experimentally intractable. The Cambridge Structural Database contains an extensive and continually expanding collection of experimentally reported metalorganic materials, while computation-ready subsets such as the CoRE-MOF databases provide thousands of structures suitable for systematic analysis. Only a small fraction (xx%) of this structural space has been examined for PFAS adsorption, however, and an even smaller number has been evaluated under environmentally rele-

[a] *Address, Address, Town, Country. Fax: XX XXXX XXXX; Tel: XX XXXX XXXX; E-mail: xxxx@aaa.bbb.ccc*
[b] *Address, Address, Town, Country.*
† Additional footnotes to the title and authors can be included *e.g.* 'Present address:' or 'These authors contributed equally to this work' as above using the symbols: ‡, §, and ¶. Please place the appropriate symbol next to the author's name and include a `\footnotetext` entry in the the correct place in the list.

vant conditions. Exhaustive synthesis and testing are therefore impractical, whereas high-level molecular simulations remain too computationally demanding to apply indiscriminately across the full database. Consequently, potentially exceptional adsorbents may remain hidden within a largely unexplored chemical space.

Machine learning offers a means of navigating this space, but its application to PFAS sorbent discovery is complicated by sparse and heterogeneous training data. Most reported studies describe materials that perform well, while rigorously defined inactive or poorly performing examples are scarce. Moreover, conventional geometric descriptorsincluding pore-limiting diameter, largest cavity diameter, accessible surface area, void fraction and framework densitycannot fully encode the chemical interactions responsible for PFAS binding. Predictive screening must therefore learn from limited labelled data while accounting for both framework geometry and chemical identity. Incorporating targeted molecular simulations into the learning cycle provides an additional route to overcome this limitation: candidates selected across the predicted performance range can be simulated, their interaction energies returned to the model, and the resulting expanded dataset used to refine subsequent predictions.

Here, we ask whether PFAS adsorbents can be prospectively identified through such an iterative, machine-learning-guided strategy. Perfluorooctanoic acid (PFOA) was selected as a prototypical target because it is a well-studied legacy C8 PFAS that combines a hydrophobic perfluoroalkyl chain with an anionic carboxylate headgroup, thereby capturing several of the competing interactions central to PFAS adsorption. We combine a curated set of MOFs with known or inferred PFOA adsorption behaviour with an unlabelled, computation-ready MOF search space. Structural descriptors calculated using Zeo++ are used to construct an initial predictive model, after which candidates spanning high-, intermediate- and low-ranking regions are subjected to molecular simulation. Iterative incorporation of the calculated PFOAframework interaction energies enables the model to evolve from an initial sparse-label classifier towards a quantitative ranking framework.

Rather than replacing simulation or experiment, this approach uses machine learning to determine where these resource-intensive methods are most informative. In doing so, it provides both a shortlist of candidate PFOA adsorbents and a chemically interpretable assessment of the framework characteristics associated with strong binding. More broadly, the study establishes a transferable route for converting crystallographic databases into actionable discovery spaces for contaminant-selective sorbents.

2 Literature Review / Data

- **Defining the chemical question and the discovery workflow**

  1. Literature-labelled active / inactive MOF space
  2. Database-mined unlabelled discovery space
  3. Relationship between Zeo++ descriptors, chemical descriptors, and adsorption behaviour

- **Building the descriptor space**

  1. Geometric descriptors retained
  2. Chemical descriptors added
  3. Why topology was considered but not pursued in the final model
  4. Descriptor sparsity, collinearity, and dimensionality constraints

3 Methodology

Will be a description of the models and how they are chained together. Novelty limited to architecture and application as opposed to model building. Based on an ML–simulation workflow starting from existing literature:

1. Baseline model (OCSVM) + binary simulation **1 iteration**
2. SVM + energy simulatio **1 iteration**
3. SVR + energy simulation **iterations**

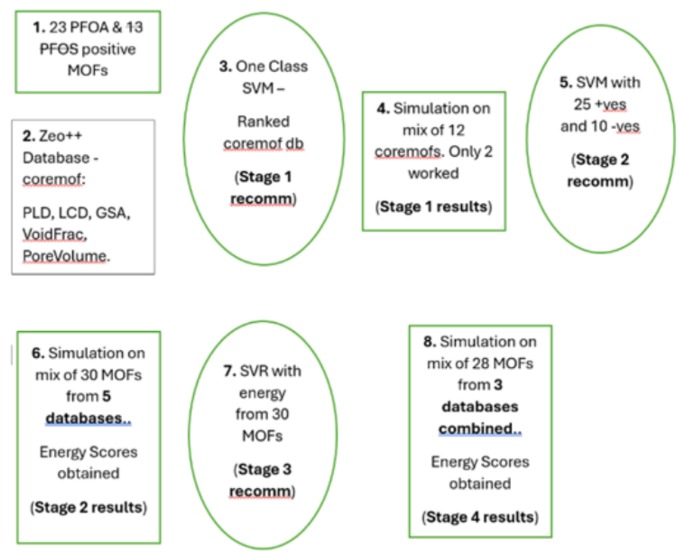

Fig. 1 Overall workflow.

3.1 Baseline Single Class Labelling model (OCSVM)

- **Establishing the baseline predictive model**

  1. Initial model trained on labelled data
  2. Positive-only or highly imbalanced learning rationale
  3. First-pass ranking of database candidates

Using the results, we are able to simulate a choice of candidates to assess **predictive power**.

The discovery of new PFAS-removing MOFs presents a challenge for conventional supervised learning approaches, as reliable negative examples are inherently difficult to define. While a subset of experimentally validated MOFs exhibiting strong PFAS adsorption performance can be identified from literature (Put original table in supporting info), the vast majority of available MOF structures lack corresponding adsorption measurements and cannot be confidently assigned as inactive. This creates a positive-unlabelled learning problem rather than a conventional binary classification task.

To address this challenge, we employ a One-Class Support Vector Machine (OC-SVM) approach, which is specifically designed to learn the characteristic feature space occupied by a single class. Unlike traditional support vector machines, which construct a maximum-margin hyperplane separating positive and negative samples, OC-SVM identifies a boundary enclosing the region of feature space associated with known positive examples. Although commonly applied for anomaly detection, including fraud identification and cybersecurity applications, the same principle can be adapted for materials discovery by defining successful PFAS-removing MOFs as the target class. The resulting model learns the structural and chemical characteristics associated with effective PFAS adsorption and enables the screening of unexplored MOFs within the CoRE MOF database prior to computationally expensive molecular simulations. (This is how the introduction finishes)

The descriptor vector of each experimentally reported PFAS-active MOF is represented as:

$$\mathbf{x}_i \in \mathbb{R}^d, \qquad i = 1, \ldots, n, \tag{1}$$

where d represents the number of physicochemical descriptors used to define the MOF feature space. Since the literature-derived dataset contains only confirmed PFAS-adsorbing MOFs, with no reliable inactive examples, the initial classification problem was formulated as a one-class learning task.

The OC-SVM attempts to separate the mapped data from the origin by finding a hyperplane:

$$\mathbf{w}^T \phi(\mathbf{x}) - \rho = 0\,, \tag{2}$$

where:

- $\phi(\mathbf{x})$ is called the kernel. It allows for the features to be transformed to a higher dimension space thus allowing non-linear separating regions (instead of hyperplanes) to enclose (rather than separate) the data. We use the RBF kernel.

- $\mathbf{w}$ are the coefficients on the now linear-in-kernel (not features) hyperplane. These can be thought of as the gradient or orienting parameters that can be manipulated to determine the separation of the hyperplane from the origin.

- ρ is the constant parameter that functions similarly in placing the hyperplane.

While the details of solving the optimisation problem that determine the best set of values for the orienting and constant parameters are not needed, we none the less briefly present the optimisation problem in order to motivate the hyper-parameters we must control. The OC-SVM is obtained by solving the following optimisation problem:

$$\begin{aligned}
&\min_{\mathbf{w},\rho,\xi} \quad \frac{1}{2}\|\mathbf{w}\|^2 + \frac{1}{\nu n}\sum_{i=1}^{n}\xi_i - \rho \\
&\text{subject to} \quad \mathbf{w}^T\phi(\mathbf{x}_i) \geq \rho - \xi_i, \qquad i = 1, \ldots, n, \\
&\qquad \xi_i \geq 0, \qquad i = 1, \ldots, n.
\end{aligned} \tag{3}$$

The minimisation across the two parameters w and ρ fits the smallest boundary encompassing the data, however the introduction of an error term ε allows fitting a tighter boundary, with the hyper-parameter ν controlling the degree to which errors may be accepted. Figure 2 show the OC-SVM results based on hyper-parameter value of $\nu = 20\%$: (figure caption needs to refer to the list of named labels).

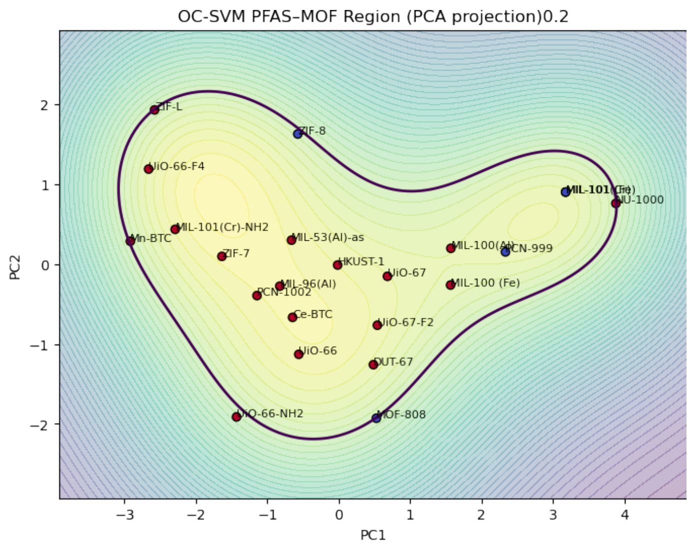

Fig. 2 OCSVM results based on starting data.

4 Conclusions

The conclusions section should come in this section at the end of the article, before the Author contributions statement and/or Conflicts of interest statement.

Author contributions

We strongly encourage authors to include author contributions and recommend using CRediT for standardised contribution descriptions. Please refer to our general author guidelines for more information about authorship.

Conflicts of interest

In accordance with our policy on Conflicts of interest please ensure that a conflicts of interest statement is included in your manuscript here. Please note that this statement is required for all submitted manuscripts. If no conflicts exist, please state that "There are no conflicts to declare".

Data availability

A data availability statement (DAS) is required to be submitted alongside all articles. Please read our full guidance on data availability statements for more details and examples of suitable statements you can use.

Acknowledgements

The acknowledgements come at the end of an article after the conclusions and before the notes and references.

Notes and references

For the reference section, the style file `rsc.bst` can be used to generate the correct reference style.§

1. Citations should appear here in the format A. Name, B. Name and C. Name, *Journal Title*, 2000, **35**, 3523;

2. A. Name, B. Name and C. Name, *Journal Title, 2000*, **35**, 3523.

...

We encourage the citation of primary research over review articles, where appropriate, in order to give credit to those who first reported a finding. Find out more about our commitments to the principles of San Francisco Declaration on Research Assessment (DORA).

1 A. Abarca, P. Gómez-Sal, A. Martín, M. Mena, J. M. Poblet and C. Yélamos, *Inorg. Chem.*, 2000, **39**, 642–651.

2 C. D. Abernethy, G. M. Codd, M. D. Spicer and M. K. Taylor, *J. Am. Chem. Soc.*, 2003, **125**, 1128–1129.

3 F. A. Cotton, G. Wilkinson, C. A. Murillio and M. Bochmann, *Advanced Inorganic Chemistry*, Wiley, Chichester, 6th edn, 1999.

4 A. J. Arduengo, III, H. V. R. Dias, R. L. Harlow and M. Kline, *J. Am. Chem. Soc.*, 1992, **114**, 5530–5534.

5 L. N. Appelhans, D. Zuccaccia, A. Kovacevic, A. R. Chianese, J. R. Miecznikowski, A. Macchioni, E. Clot, O. Eisenstein and R. H. Crabtree, *J. Am. Chem. Soc.*, 2005, **127**, 16299–16311.